Pipeline Internals

Pipeline Internals I

This supplemental section documents the data structures and on-disk artifacts the pipeline produces, for readers who want to extend or audit it.

The Mermaid class diagram in this subsection shows the canonical fields each record carries through the pipeline. Records have additional optional metadata (e.g. `Paper.url`, `Paper.publisher`, `Paper.isbn`, `Paper.raw`) omitted for readability — consult `infrastructure/search/literature/models.py` (source on GitHub) and `src/pipeline/pipeline.py` (source on GitHub) for the full schema.

Data structures I

```
classDiagram
    class SearchQuery {
        +str text
        +int max_results
        +int|None year_min
        +int|None year_max
        +list~str~ sources
        +list~str~ fields
        +matches_year(year) bool
    }

    class Paper {
        +str id
        +str title
        +list~str~ authors
        +int|None year
        +str|None doi
```

Data structures II

```
+str|None url
+str|None abstract
+str|None pdf_url
+str|None fulltext
+str|None venue
+str|None venue_type
+str|None volume
+str|None issue
+str|None pages
+str|None publisher
+str|None edition
+str|None isbn
+list~str~ keywords
+str|None source
+float score
+dict raw
+to_dict() dict
+from_dict(d) Paper
```

Data structures III

```
        }

        class SearchResult {
            +SearchQuery query
            +list~Paper~ papers
            +dict per_source_counts
            +dict errors
            +to_dict() dict
            +to_json() str
        }

        class BibEntry {
            +str entry_type
            +str citation_key
            +OrderedDict fields
        }

        class BibDatabase {
```

Data structures IV

```
        +list~BibEntry~ entries
        +str|None preamble
        +find(key) BibEntry
    }

    class LiteratureRunArtifacts {
        +SearchResult result
        +Path|None corpus_path
        +Path|None bibtex_path
        +list~FetchResult~ enrichment_log
        +Path|None cache_dir
        +dict citation_keys
        +papers() list~Paper~
    }

    class ManuscriptVariables {
        +str config_query
        +int config_max_results
```

Data structures V

```
        +str config_sources
        +int result_num_papers
        +int result_num_sources
        +str result_per_source
        +str result_errors
        +str result_year_min
        +str result_year_max
        +int result_with_abstract
        +int result_with_doi
        +int deep_max_results_per_keyword
        +int deep_keyword_count
        +str deep_keywords_joined
        +str deep_sources
        +str deep_unique_papers
        +as_dict() dict
        +as_uppercase_keys() dict
    }
```

Data structures VI

```
class SynthesisResult {
    +str kind
    +str prompt
    +str text
    +str|None paper_id
}

SearchResult --> SearchQuery
SearchResult --> "*" Paper
LiteratureRunArtifacts --> SearchResult
LiteratureRunArtifacts --> "*" BibEntry : via
paper_to_bibentry
BibDatabase --> "*" BibEntry
```

On-disk layout I

The project keeps committed input data in `data/`, regeneratable outputs in `output/` (gitignored), and the manuscript source in `manuscript/`. The Mermaid flowchart in this subsection illustrates every artefact the standard pipeline writes:

On-disk layout I

- `data/corpus.json` — the bundled offline corpus, CI-safe default for `LocalBackend`.
- `output/search/results.json` — raw `SearchResult` JSON from the latest run.
- `output/search/cache/search_<hash>.json` — deterministic `SearchCache` files.
- `output/cache/abs/<safe_id>.txt` — one cached abstract per paper.
- `output/cache/pdf/<safe_id>.{pdf,txt}` — PDF bytes plus extracted text.
- `output/llm/synthesis.md` — corpus-level LLM synthesis (when enabled).
- `output/llm/per_paper/<safe_id>.md` — one per-paper note per paper (when enabled).
- `output/figures/{papers_per_source,year_histogram,score_distributio` — diagnostic figures.
- `output/data/manuscript_variables.json` — substitution table consumed by the resolver.
- `output/corpus.json` — enriched corpus, written in `LocalBackend`-compatible format so it can re-seed a deterministic future run.

On-disk layout II

- `output/enrichment_log.json` — one entry per fetcher per paper.
- `output/reading_report.md` — final markdown reading report.
- `output/run_summary.json` — one-line metadata for the run.
- `manuscript/references.bib` — auto-populated, Pandoc-ready BibTeX.

```
flowchart TB
    P["projects/templates/template_search_project/"]
    P --> DAT["data/<br/>committed"]
    P --> OUT["output/<br/>regeneratable · gitignored"]
    P --> MAN["manuscript/"]

    DAT --> CORP_IN[corpus.json<br/>bundled offline corpus<br/>CI-safe default]

    OUT --> SR["search/"]
    OUT --> CC["cache/"]
    OUT --> LD["llm/"]
    OUT --> FIG["figures/"]
```

On-disk layout III

```
OUT --> DD["data/"]
OUT --> CORP_OUT[corpus.json<br/>enriched · LocalBackend-compatible]
OUT --> ELOG[enrichment_log.json<br/>per-fetch status]
OUT --> RR[reading_report.md]
OUT --> RS[run_summary.json]

SR --> SR_R[results.json<br/>SearchResult]
SR -->
SR_C["cache/search_HASH.json<br/>deterministic SearchCache"]

CC --> CC_A["abs/SAFE_ID.txt"]
CC --> CC_P["pdf/SAFE_ID.pdf and .txt"]

LD --> LD_S[synthesis.md]
LD --> LD_PP["per_paper/SAFE_ID.md"]
```

On-disk layout IV

```
FIG --> F1[papers_per_source.png]
FIG --> F2[year_histogram.png]
FIG --> F3[score_distribution.png]

DD --> MV[manuscript_variables.json]

MAN --> BIB[references.bib<br/>auto-populated · Pandoc-ready]

classDef dir fill:#0f172a,stroke:#0f172a,color:#fff
classDef gen fill:#0f766e,stroke:#0f172a,color:#fff
classDef src fill:#1e3a8a,stroke:#0f172a,color:#fff
class P,DAT,OUT,MAN,SR,CC,LD,FIG,DD dir
class CORP_OUT,ELOG,RR,RS,SR_R,SR_C,CC_A,CC_P,LD_S,LD_PP,F1,F2,F3,MV,BIB gen
```

On-disk layout V

```
class CORP_IN src
```

Citation-key collision handling I

`paper_to_bibentry()` generates citation keys as `<author><year><title-word>` (with stop-words filtered and unicode folded). When two papers in the same result set produce the same key — common when one author publishes multiple papers in the same year on closely related topics — `src/pipeline/pipeline.py::_disambiguate_citation_key` appends a deterministic suffix from the alphabet (`a`, `b`, …, `z`, then two-letter combinations `aa`, `ab`, …) until uniqueness is restored, with a numeric `_1`, `_2`, … fallback for the pathological case. The mapping is exposed to downstream stages via `LiteratureRunArtifacts.citation_keys`, and the report uses these keys verbatim, so the LLM synthesis and the BibTeX file always agree.

Citation-key collision handling I

The deep-search workflow has its own collision handler in `src/search/deep_search.py::run_deep_search` that operates over the post-deduplication aggregate roster — see sec. 8 — and the unified `references_deep.bib` reflects the same mapping.

Failure isolation I

- **A backend is unreachable.** `LiteratureClient` records the message into `result.errors[name]` and continues. The reading report surfaces these errors in a callout block at the top.
- **An abstract fetch fails.** The fetcher records `status="error"` in `enrichment_log.json` and the paper keeps its existing (possibly empty) abstract.
- **A PDF fetch fails or `pypdf` is missing.** Same pattern — the PDF, if downloaded, is still cached on disk. The reading report does not reference the missing fulltext.
- **The LLM is unreachable or unconfigured.** `scripts/run_search_pipeline.py` logs a warning and skips the synthesis stage entirely — `output/llm/` is left empty and the reading report omits both the per-paper-notes and cross-corpus sections. No placeholder text is ever written into the archive, so a missing LLM is observable from the absence of those sections rather than from a fake "(LLM unavailable)" string.